

ROTATIONAL EFFECTS IN COMPLEX-FORMING BIMOLECULAR REACTIONS: APPLICATION TO THE REACTION $\text{CH}_4 + \text{O}_2^+ *$

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ABSTRACT

A statistical theory of complex-forming bimolecular reactions, involving loose entrance and rigid exit activated complexes, is elaborated and analyzed with respect to rotational effects. The reaction $\text{CH}_4 + \text{O}_2^+$ serves as an illustrative example. First, the different rotational dependences of the entrance and exit threshold energies can lead to “rotational channel switching”, i.e. change of the order of the threshold energies with rotation. Second, rotational effects in the numbers of open channels for forward and backward dissociation of the complex lead to energy- and rotation-dependent yields of the reaction. Finally, nuclear spin statistical weights of rotational levels at low temperatures can lead to very specific isotope effects. The experimental results for $\text{CH}_4 + \text{O}_2^+$ can well be reproduced. However, because of the uncertainties in the properties of the rigid exit activated complex, this fit cannot be made in a unique way.

INTRODUCTION

Bimolecular reactions involving the formation of intermediate bound complexes



have interesting properties. They may show temperature dependences which strongly deviate from Arrhenius expressions. They may also be pressure-dependent as well as dependent on the nature of the bath gas when collisional energy transfer interferes with the steps shown in Eq. (1).

A formal representation of the overall rate coefficient, k , of reaction (1) leads to

$$k \approx k_1 \left\{ \frac{k_2}{k_{-1} + k_2} \right\} \quad (2)$$

* Dedicated to E.E. Ferguson in honour of his contributions to ionic physics and chemistry.

However, factoring of the rate coefficient, k , into a capture part k_1 and a yield factor $k_2/(k_{-1} + k_2)$ generally will not be possible because of the different energy dependences of the specific rate constants k_{-1} and k_2 for backward and forward unimolecular dissociation of the complex AB^* . The rate coefficient, k_1 , for complex formation is directly linked with k_{-1} via detailed balancing such that the problem can be entirely formulated in the framework of unimolecular rate theory. Besides the energy dependence of k_1 , k_{-1} , and k_2 , the angular momentum dependence has also to be taken into account. Considering the formation of complexes AB^* with a given energy, E , and total angular momentum with quantum number, J , Eq. (2) has to be replaced by a suitable thermal average

$$k(T) = \left\langle k_1(E, J) \left\{ \frac{k_2(E, J)}{k_{-1}(E, J) + k_2(E, J)} \right\} \right\rangle_T \quad (3)$$

Detailed formulations of this type have been elaborated in various ways, e.g. by phase space theory, Miller's unified statistical model, the statistical adiabatic channel model, and others (for a comparison of these approaches and the relevant references, see ref. 1).

In the present work, we shall employ the statistical adiabatic channel model (SACM). In this general statistical theory [1-3], Eq. (3) is transformed into

$$k = \frac{kT}{h} \left(\frac{Q_{AB}^*}{Q_A Q_B} \right)_{\text{el,trans}} \left(\frac{1}{Q_A Q_B} \right)_{\text{vib,rot}} \times \sum_{J=0}^{\infty} (2J+1) \int_0^{\infty} \frac{dE}{kT} \exp\left(-\frac{E}{kT}\right) \left[\frac{W_a(E, J) W_b(E, J)}{W_a(E, J) + W_b(E, J)} \right] \quad (4)$$

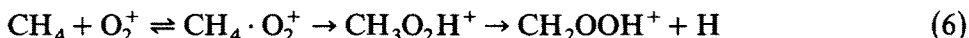
where Q denotes partition functions for electronic, translational, vibrational, and rotational states, respectively, $W_a(E, J)$ is the number of open channels for dissociation of the complex AB^* backward into $A + B$ (corresponding to k_{-1}), $W_b(E, J)$ is the number of open channels for forward reaction (corresponding to k_2). Equation (4) includes collisional energy transfer involving complex formation when A and C are different states of A and $B = D$ is a bath gas. Equation (4) also includes capture or high-pressure association processes, when $W_b(E, J) \gg W_a(E, J)$ such that the term in brackets in Eq. (4) is reduced to $W_a(E, J)$.

Equation (4) has only rarely been evaluated explicitly, in particular with respect to the rotational effects. In one previous treatment [4], we have demonstrated the importance of these effects for the reaction



which involves the intermediate formation of HO_2^* complexes. Different reduced moments of inertia of reactants and products here lead to marked differences in the specific rate constants $k(E, J)$ for forward and backward dissociation of the complex. As a consequence, in spite of the 70 kJ mol^{-1} exothermicity of reaction (5), there is considerable redissociation $\text{HO}_2^* \rightarrow \text{HO} + \text{O}$ which gains importance with increasing temperature.

Rotational effects in Eq. (4) can be of various types: (i) depending on the complexity (atomic, linear, non-linear, etc.) of the species A–D and their rotational constants, the numbers of open channels $W_a(E, J)$ and $W_b(E, J)$ can have quite different forms; (ii) the threshold energies $E_{\text{oa}}(J)$ and $E_{\text{ob}}(J)$ for backward and forward dissociation of the complex AB^* can vary with J such that even “channel switching” occurs [5,6] (i.e. the ordering of threshold energies may change with J); (iii) for species with identical atoms, nuclear spin statistics can introduce selective populations of rotational states which, for low temperatures, specifically influence k and lead to interesting isotope effects. Having investigated some of these contributions in the reaction (5), where complex formation and dissociation are both characterized by relatively “loose activated complexes”, we would also like to demonstrate rotational effects for a loose entrance/rigid exit situation. According to the extensive work by Ferguson and his colleagues in several countries [7], such an example is provided by the ion/molecule reaction



The final loss of a hydrogen atom is a rapid process such that Eq. (4) applies well to the first steps of the reaction.

A first analysis of the experimental results on reaction (6) has already been given in ref. 7 on the basis of different energy dependences of k_{-1} and k_2 . Since backward dissociation (k_{-1}) involves a loose activated complex, whereas the forward isomerization proceeds via a rigid complex, k_{-1} is much larger and has a much stronger energy dependence than k_2 , such that $k_2/(k_{-1} + k_2)$ is a function which strongly decays with increasing energy (or temperature). Rotational effects in ref. 7 were only included implicitly by identifying k_1 in Eq. (1) with the Langevin capture rate coefficient and by representing $k_{-1}(E)$ by phase space theory. The present work adopts a more detailed point of view, demonstrating the origin of various rotational contributions to the rate coefficients. Unfortunately, not all ingredients of the theory are presently accessible such as, in particular, the height of the energy barrier for isomerization $\text{CH}_4 \cdot \text{O}_2^+ \rightarrow \text{CH}_3\text{O}_2\text{H}^+$ and other properties of the corresponding activated complex. As a consequence, the measured rate coefficients can be reproduced in various ways, which prevents a unique determination of the isomerization barrier from the experiments. Therefore, in this respect, our treatment will remain more on a demonstrative level. On

the other hand, the capture process $\text{CH}_4 + \text{O}_2^+$ appears to be fully characterized by an isotropic polarization potential such that phase space theory [8,9] can be applied, simplifying the calculation of $k_1(E, J)$ and $k_{-1}(E, J)$. In the case of anisotropic potentials, phase space theory would not be valid and particular SACM-modifications would have to be introduced [10–12].

ROTATIONAL EFFECTS ON THRESHOLD ENERGIES

At first, we consider the rotational dependence of the threshold energies $E_{\text{oa}}(J)$ and $E_{\text{ob}}(J)$ for backward and forward reaction of the complex AB^* . We specifically analyze the reaction $\text{CH}_4 + \text{O}_2^+$ for which Fig. 1 sketches a potential diagram [7]. Representing the $\text{CH}_4 + \text{O}_2^+$ capture by an isotropic polarization potential, one can easily derive the threshold energy $E_{\text{oa}}(l)$, where l is the quantum number for orbital rotation. Since the threshold energies $E_{\text{oa}}(l)$ are very low, collisions with a large range of l values contribute. Orbital angular momentum (l) combines with rotational angular momenta of CH_4 and O_2^+ (j_1 and j_2) to give the total angular momentum (J). Because of the relatively large rotational constants of CH_4 and O_2^+ , for the majority of contributing channels, $l \gg j_1, j_2$ and $J \approx l$ can approximately be used. Therefore, $E_{\text{oa}}(l) \approx E_{\text{oa}}(J)$ leads to the relationship (see, for example, refs. 11 and 12)

$$E_{\text{oa}}(J) \approx C_a [J(J+1)]^2 \quad (7)$$

with

$$C_a = \frac{\hbar^4}{8\mu^2 \alpha q^2} \quad (8)$$

Here, q is the ionic charge of O_2^+ , α is the isotropic polarizability of CH_4 , and μ is the reduced mass of the collision pair. In the presence of an

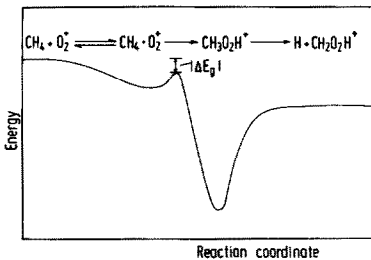


Fig. 1. Energy profile of the reaction $\text{CH}_4 + \text{O}_2^+ \rightarrow \text{CH}_3\text{O}_2\text{H}^+$ analyzed in this work (mechanism adopted from ref. 7).

anisotropy of the potential, the zero-point energy of the hindered rotational motion in the anisotropic potential would have to be added to Eq. (7) (see ref. 12). The threshold energy, $E_{ob}(J)$, of the isomerization $\text{CH}_4 \cdot \text{O}_2^+ \rightarrow \text{CH}_3\text{O}_2\text{H}^+$ is not known. We represent this by

$$E_{ob}(J) \approx -|\Delta E_g| + C_b J(J+1) \quad (9)$$

where C_b denotes the rotational constant of the probably rigid activated complex, being close to the rotational constant of the collision complex $\text{CH}_4 \cdot \text{O}_2^+$. The high value of the experimental rate coefficient k for $T \rightarrow 0$ K suggests $E_{ob}(J=0) < E_{oa}(J=0)$. Nevertheless, $E_{ob}(J) > E_{oa}(J)$ can occur at higher J if $C_a < C_b$. Figure 2 demonstrates such rotational channel switching for the two-channel reaction of the complex $\text{CH}_4 \cdot \text{O}_2^+$. The exact J values where the curves cross, of course, depend on the unknown value of $|\Delta E_g|$. (The input data of Fig. 2 are $\alpha = 2.6 \text{ \AA}^3$, $\mu = 1.77 \times 10^{-23} \text{ g}$, $C_a = 4.1 \times 10^{-6} \text{ cm}^{-1}$ and $C_b \approx 0.4 \text{ cm}^{-1}$ by analogy with related molecules [13]).

Neglecting any other rotational effects, the different J dependence of the threshold energies $E_{oa}(J)$ and $E_{ob}(J)$ can alone lead to a negative temperature coefficient of k , in contrast to the temperature-independent Langevin capture rate coefficient k_L given by

$$k_L = 2\pi q \sqrt{\frac{\alpha}{\mu}} \quad (10)$$

For illustrative purposes, we calculate k from Eq. (4) with a yield factor $W_b(E, J)/[W_a(E, J) + W_b(E, J)]$ equal to unity for $E > E_{ob}(J)$ and equal to zero for $E < E_{ob}(J)$. Then, $\sum_J \int_0^\infty dE$ becomes equal to the product

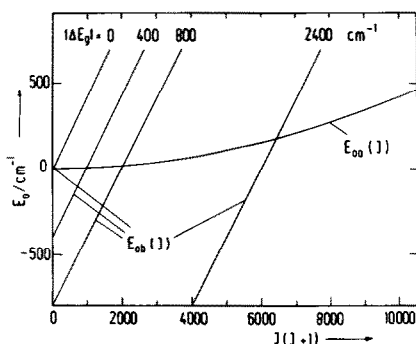


Fig. 2. Rotational channel switching: channel threshold energies $E_{oa}(J)$ for $\text{CH}_4 + \text{O}_2^+ \rightarrow \text{CH}_4 \cdot \text{O}_2^+$, and $E_{ob}(J)$ for $\text{CH}_4 \cdot \text{O}_2^+ \rightarrow \text{CH}_3\text{O}_2\text{H}^+$ (molecular parameters given in the text; for $|\Delta E_g|$, see Fig. 1).

of $(Q_A Q_B)_{\text{vib,rot}}$ and the centrifugal partition function, Q_{cent}^* , which is given by

$$Q_{\text{cent}}^* = \sum_{J=0} (2J+1) \exp\left(-\frac{E_0^*(J)}{kT}\right) \quad (11)$$

Here, $E_0^*(J) = \max [E_{0a}(J), E_{0b}(J)]$. If $E_0^*(J) = E_{0a}(J)$, one would have

$$Q_{\text{cent}, L}^* = \sqrt{\frac{2\pi\alpha kT\mu^2 q^2}{\hbar^4}} \quad (12)$$

such as used [11,12] for the derivation of the Langevin rate constant given by Eq. (10). With Eqs. (7)–(9) one obtains for the hypothetical ratio

$$\frac{k}{k_L} = \frac{Q_{\text{cent}}^*}{Q_{\text{cent}, L}^*} \quad (13)$$

the result

$$\begin{aligned} \frac{k}{k_L} = & \operatorname{erf}\left\{J_1(J_1+1)\sqrt{\frac{C_a}{kT}}\right\} + 1 - \operatorname{erf}\left\{J_2(J_2+1)\sqrt{\frac{C_a}{kT}}\right\} \\ & + \sqrt{\frac{4C_a kT}{\pi C_b^2}} \left[\exp\left\{\frac{-J_1(J_1+1)C_b}{kT}\right\} - \exp\left\{\frac{-J_2(J_2+1)C_b}{kT}\right\} \right] \end{aligned} \quad (14)$$

J_1 and J_2 denote the lower and upper J values, respectively, for which $E_{0a}(J) = E_{0b}(J)$. Equations (7)–(9) indicate how J_1 and J_2 depend on $|\Delta E_g|$. Figure 3 shows the results for various values of $|\Delta E_g|$. The figure illustrates that “rotational channel switching” may be one of the reasons for the negative temperature coefficient of k .

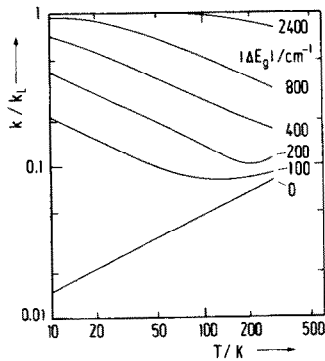


Fig. 3. Rotational channel switching: rate coefficients, k , in the absence of redissociation (i.e. $W_b \gg W_a$ at $E > E_{0b}$, k_L = Langevin rate coefficient; for $|\Delta E_g|$, see Fig. 1).

ROTATIONAL EFFECTS ON NUMBERS OF OPEN CHANNELS

In this section, we consider the energy and angular momentum dependences of the numbers of open channels $W_a(E, J)$ and $W_b(E, J)$. We illustrate these effects by showing the yield ratios $W_b(E, J)/[W_a(E, J) + W_b(E, J)]$. At this stage, we neglect the problem of nuclear spin statistics, symmetry numbers, and reaction path degeneracy factors. We assume that the formation of the $\text{CH}_4 \cdot \text{O}_2^+$ complex is completely governed by an isotropic polarization potential such that the SACM treatment coincides with phase space theory [8,9]. On the other hand, we assume that the isomerization $\text{CH}_4\text{O}_2^+ \rightarrow \text{CH}_3\text{O}_2\text{H}^+$ involves a rigid activated complex such that the SACM treatment coincides with rigid RRKM theory.

The association $\text{CH}_4 + \text{O}_2^+$ corresponds to a spherical top + linear molecule system, for which phase space theory gives [9]

$$W_a(E^*, J) \approx \frac{(2J+1)E^{*2}}{2B_{\text{CH}_4}^{3/2}B_{\text{O}_2}^{1/2}} \arcsin \sqrt{\frac{B_{\text{CH}_4}}{B_{\text{CH}_4} + B_{\text{O}_2}}} \quad (15)$$

for small J and

$$W_a(E^*, J) \approx \frac{8 E^{*5/2}}{15 B_{\text{CH}_4}^{3/2} B_{\text{O}_2}} \quad (16)$$

for large J . Here E^* indicates the energy of the complex in excess of $E_{\text{oa}}(J)$ and B denotes the respective rotational constants. In ref. 7, apart from symmetry numbers, $W_a(E) = E^2/2B_{\text{CH}_4}B_{\text{O}_2}$ was taken from ref. 14. This expression apparently corresponds to $J = 0$, but the detailed value is different from Eq. (15) because of the violation of angular momentum coupling restrictions in the derivation in ref. 14. For intermediate J values, there is a transition between Eqs. (15) and (16) which is difficult to represent by an analytical formula such that explicit state counting is required. However, always taking the smaller of the two limiting expressions, i.e.

$$W_a(E^*, J) = \min\{W_a[\text{Eq. (15)}], W_a[\text{Eq. (16)}]\} \quad (17)$$

gives very satisfactory results (see the explicit expressions of ref. 9 for the atom + spherical top). Equations (15)–(17) so far have not included contributions of the vibrations of CH_4 and O_2^+ . After thermal averaging, apart from the yield ratio, this contribution cancels against $(Q_A Q_B)_{\text{vib}}$ in Eq. (4). Furthermore, the vibrational quanta of CH_4 and O_2^+ are so large that their contributions to the yield factor remain negligible for the considered temperature range, i.e. $T < 600$ K.

Rotational effects on numbers of open channels $W_b(E, J)$ for rigid activated complexes have only rarely been demonstrated explicitly (see the

example of H_2CO in ref. 6). Treating the activated complex as a symmetric top with rotational constants A^* and B^* , one would obtain [6]

$$W_b(E, J) = \sum_{K=-J}^J W^* [E - E_{\text{ob}}(J) - (A^* - B^*)K^2] \quad (18)$$

where W^* is obtained by Beyer-Swinehart state counting using the adequate harmonic oscillator frequencies of the complex. Since only small energy ranges are of interest here, high-frequency modes do not contribute to W^* . One can only speculate whether low-frequency modes play a role or not. To a first approximation, we assume that there are no activated complex modes which have sufficiently low frequencies to contribute in the relevant energy range. The contribution of the K rotor depends on A^* and B^* . As before, we use $B^* \approx 0.4 \text{ cm}^{-1}$; we also estimate $A^* \approx 5 \text{ cm}^{-1}$ by analogy with related molecules [13] such as H_3CCl . For the given activated complex model, $W_b(E, J)$ follows from Eq. (18) as

$$W_b(E, J) \approx \min\{(2J+1), (2J_3+1)\} \quad (19)$$

where

$$J_3 = \sqrt{\frac{[E - E_{\text{ob}}(J)]}{(A^* - B^*)}}$$

It would be easy to consider excitation of vibrations. For example, if one quantum, $h\nu$, of a low frequency vibration could be excited, $W_b(E, J)$ would be given by the sum of Eq. (19) and the corresponding value with E replaced by $E - h\nu$.

Equations (15) and (16) are classical expressions becoming invalid at $E \leq 2B$. They also neglect symmetry and nuclear spin statistics. Such effects will be considered in the next section. For the moment, we exclude very low temperatures where individual rotational levels have to be inspected with respect to their symmetry properties. Instead, we take this effect into account in an average way by dividing $W_a(E, J)$ by a symmetry number $\sigma(\text{CH}_4)\sigma(\text{O}_2^+) = 12 \times 2$ and omitting spin statistical weights. In doing this, there are no further reaction path degeneracies [1,2]. Symmetry numbers and reaction path degeneracies in the past have often been mixed in an inadequate way. In the absence of information on the structure of the isomerization activated complex, we tentatively divide $W_b(E, J)$ by a symmetry number of 3, at least as long as $W_b \gg j$.

In order to demonstrate rotational effects on numbers of open channels, at first, in Fig. 4, we show a yield ratio

$$Y = \frac{W_b(E, J)}{W_a(E, J) + W_b(E, J)}$$

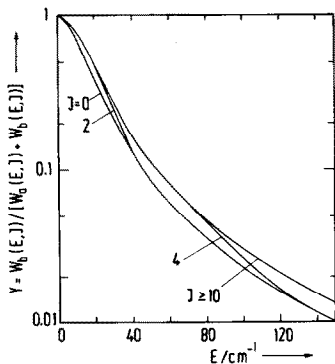


Fig. 4. Yield ratio $Y(E, J)$ neglecting rotational channel switching [i.e. $E_{\text{oa}} = E_{\text{ob}} = 0$, classical expressions Eqs. (15)–(19), see text].

neglecting the dependence of the threshold energies E_{oa} and E_{ob} on J . [Equations (15)–(19) were used with $B_{\text{CH}_4} = 5.24 \text{ cm}^{-1}$, $B_{\text{O}_2^+} = 1.67 \text{ cm}^{-1}$, $E_{\text{oa}} = E_{\text{ob}} = 0$, $\sigma(\text{CH}_4)\sigma(\text{O}_2^+) = 24$, $\sigma(\text{CH}_4\text{O}_2^+ \rightarrow \text{CH}_3\text{O}_2\text{H}^+) = 3$.] Y markedly decreases with increasing energy. There is some J dependence because, in certain ranges, W_a and W_b are proportional to $(2J + 1)$. However, much of this dependence cancels out. Since large ranges of J values contribute to the rate coefficient, the $J = 10$ curve would provide a sufficient representation for all J values. This curve corresponds to $Y \approx [1 + (E/18 \text{ cm}^{-1})^2]^{-1}$.

Obviously, the decrease of the yield ratio, Y , with increasing energy constitutes a major contribution to the negative temperature coefficient of the rate coefficient k . Figure 4 only shows a minor J dependence. However, this impression is misleading. When the neglected J dependence of the threshold energies is also taken into account, the J dependence of the yield ratio intensifies since there is less cancelling of the factors $2J + 1$ in W_a and W_b . A comparison of the experimental results [7], i.e. $k/k_L \approx 0.4$ at 20 K and 0.025 at 100 K, with Fig. 3 shows that $|\Delta E_g|$ must be larger than 300 cm^{-1} since rotational channel switching and yield ratios, Y , smaller than unity both reduce k/k_L .

Choosing a very large $|\Delta E_g|$ value such that Eq. (19) is always equal to $2J + 1$, one would obtain a much stronger J dependence, such as demonstrated in Fig. 5. However, if $|\Delta E_g|$ were so large that Eq. (19) would always be characterized by $2J + 1$, vibrational contributions in $W_b(E, J)$ would certainly have to be included with the effect of an even weaker decay of Y with decreasing energy. This would result in a too weak temperature coefficient of k . Therefore, the rotational dependences of the numbers of

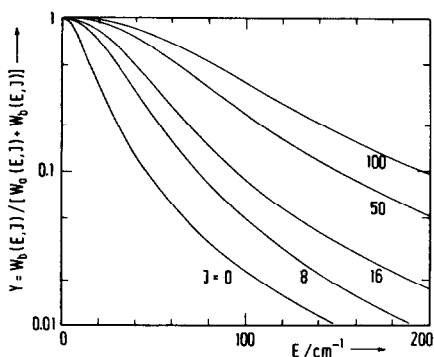


Fig. 5. Yield ratio $Y(E, J)$ taking into account rotational channel switching [large $|\Delta E_g|$ such that $W_b = 2J + 1$ in Eq. (19)].

open channels and of the threshold energies have to be considered simultaneously.

Figure 6 shows the yield ratio $Y(E, J)$ calculated simultaneously taking into account the J dependences of threshold energies and numbers of open channels, assuming a $|\Delta E_g|$ value of 400 cm^{-1} . The effect of this $|\Delta E_g|$ value is two-fold: there is rotational channel switching leading to $Y = 0$ at small energies for $J \geq 32$; there is also a limitation of the increase of Y with increasing J (at $J \geq 8$) due to a replacing of $2J + 1$ by $2J_3 + 1$ in Eq. (19). The yield ratio decreases to very small values soon after J exceeds 31, such that higher J values hardly contribute at all to the reaction, even if the energy exceeds $E_o(J)$. This cut-off value of J corresponds to $E_{oa}(J) \approx E_{ob}(J)$, i.e. $C_b J(J + 1) \approx |\Delta E_g|$ [because $E_{oa}(J) \approx 0$].

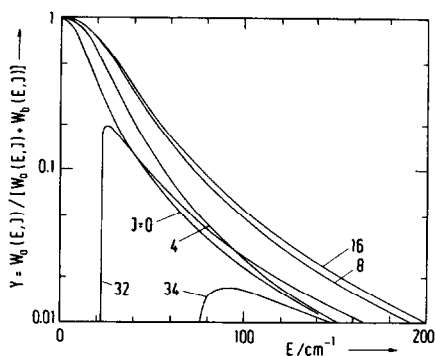


Fig. 6. Yield ratio $Y(E, J)$ taking into account rotational channel switching ($|\Delta E_g| = 400 \text{ cm}^{-1}$).

EFFECTS OF NUCLEAR SPIN STATISTICS

The statistical weights of the rotational states of CH_4 and O_2^+ , as well as of the adiabatic channels correlating with these states, are given by symmetry properties and nuclear spin statistics. The same applies to the channels leading to $\text{CH}_3\text{O}_2\text{H}^+$. For sufficiently large numbers of channels, these effects on the average are accounted for by dividing the “mechanical” $W(E, J)$ by symmetry numbers [1,2] and multiplying by overall spin statistical weights. At small excitations of the rotations of CH_4 and O_2^+ , this procedure is insufficient and the weights of the rotational levels have to be considered individually. On the other hand, the large range of total J values produced by the large relevant range of l , except for the smallest J , allows for an average treatment of the $\text{CH}_3\text{O}_2\text{H}^+$ rotational states.

The statistical weights of the first 16 rotational levels of tetrahedral molecules such as CH_4 and CD_4 are tabulated in ref. 15. For $\text{O}_2^+(\text{}^2\Pi_g)$, in principle Λ -doubling should also be taken into account [16]. However, since the electronic properties of the complex are not known (or directly related to the $\text{}^2\Pi_g$ state), we omit the doubling and simply leave out every second rotational state of O_2^+ . The determination of $W_a(E^*, J)$ in phase space theory, neglecting vibrational contributions, is based on the summation [9]

$$W_a(E^*, J) = \sum_{j_1} g(j_1) \sum_{j_2} g(j_2) \sum_j \sum_l 1 \quad (20)$$

with the constraints $0 \leq B_1 j_1(j_1 + 1) \leq E^*$, $0 \leq B_2 j_2(j_2 + 1) \leq E^* - B_1 j_1(j_1 + 1)$, $|j_1 - j_2| \leq j \leq j_1 + j_2$, and $|J - j| \leq l \leq J + j$ and the relevant statistical weights $g(j_1)$ and $g(j_2)$. For O_2^+ , $g(j_2)$ alternates between 1 and 0. For CH_4 , we use $g(j_1) = 5, 3, 5, 11, 13, 11, 21, 19, \dots$ for $j_1 = 0, 1, 2, 3, 4, 5, 6, 7, \dots$. For CD_4 , one would have $g(j_1) = 15, 18, 30, 51, 63, 66, 96, 99, \dots$ for $j_1 = 0, 1, 2, 3, 4, 5, 6, 7, \dots$. The rotational partition functions of CH_4 , CD_4 , and O_2^+ also have to account for symmetry and nuclear spins. They are counted for low temperatures or, for high temperatures, they are calculated classically, divided by the symmetry numbers and multiplied by the spin statistical weight $(2I + 1)^4$. Figure 7 compares $W_a(E^*, J)$, such as calculated via Eq. (20), with Eqs. (15)–(17) divided by $\sigma(\text{CH}_4)\sigma(\text{O}_2^+) = 24$ and multiplied by $(2I + 1)^4 = 16$. For sufficiently high energies $E^* = E - E_{0a}$, the two expressions approach each other. However, at low energies, individual statistical weights have some influence. It should also be noted that, at low energies, it matters whether even or odd rotational states of O_2^+ are included, see Fig. 7.

The accurate determination of the numbers of open channels for the rigid activated complex of the process $\text{CH}_4 \cdot \text{O}_2^+ \rightarrow \text{CH}_3\text{O}_2\text{H}^+$ presently appears to be impossible. For illustrative purposes, we assume that the complex has

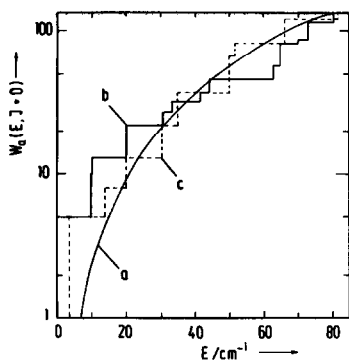


Fig. 7. Number of open channels $W_a(E, J) = 0$ accounting for nuclear spin statistical weights. Curve a: classical expression Eq. (15) multiplied by $(2I+1)^4/\sigma(\text{CH}_4)\sigma(\text{O}_2^+) = 16/24$; curve b: counted expression Eq. (20), even levels of O_2^+ included; curve c: as curve b, odd levels of O_2^+ included.

a three-fold symmetry. For a three-fold symmetry, treating the statistical weights of the three H atoms of CH_3 and of the fourth H atom in a separate way leads to statistical weights $(2I+1)^2(4I^2+4I+3)/3$ for K divisible by 3 (or $K=0$), and $(2I+1)^2(4I^2+4I)/3$ for K not divisible by 3. At sufficiently large K values, these weights, on average, lead to a factor of $(2I+1)^4/\sigma = 16/3$ to be multiplied with the classical value of $W_b(E, J)$. For sufficiently large values of W_a and W_b , the nuclear spin factors $(2I+1)^4$ cancel from the expression for k and can be omitted completely.

THERMAL RATE COEFFICIENTS

In the foregoing sections, we have illustrated the various ways by which rotational effects enter the calculation of the rate coefficient, k , for a "loose entrance/rigid exit" bimolecular reaction involving complex formation. Thermal averaging of the derived results directly leads to the thermal rate coefficients, k . A comparison with the experiments clearly indicates that rotational effects alone already require a relatively large $|\Delta E_g|$ value such as defined in Fig. 1. On the other hand, if $|\Delta E_g|$ is of the order of (or larger than) 400 cm^{-1} , vibrational contributions to $W_b(E, J)$ of the rigid activated complex at the exit certainly come into play, since one is no longer close to the threshold of the exit activated complex where vibrational excitation could have been neglected. Unfortunately, apart from the symmetry uncertainties of this activated complex, there is complete uncertainty about the low-frequency modes of this complex. The net result of the vibrations is an increase of $W_b(E, J)$ with increasing energy, which results in an increase of $Y(E, J)$ compared with the vibrationless case. As demonstrated in ref. 7,

this may produce a change of the temperature coefficient of k from negative to positive values, such as is observed experimentally at $T \approx 300$ K. By changing the vibrational frequencies of the complex, nearly any experimental temperature dependence can be reproduced. Because of this present uncertainty, in the following we do not proceed toward an accurate thermal averaging of our previous results considering $|\Delta E_g|$ as an adjustable parameter. Instead, we use approximate averaging techniques, which nevertheless reveal the essential consequences of the rotational effects.

We only consider the range $T < 300$ K with negative temperature coefficients of k . Here, the experimental data can well be represented by $k/k_L \approx [1 + (T/16 \text{ K})^2]^{-1}$. Thermal averaging over the yield ratios $Y(E, J)$, taking into account J -dependent threshold energies, can be approximated in the following way. According to Fig. 6, the yield ratios soon become very small when J exceeds $J_1 \approx \sqrt{|\Delta E_g|/0.4 \text{ cm}^{-1}}$ from Eq. (9). Therefore, we omit higher J values from the sum in Eq. (4). Furthermore, with increasing J , the yield ratio, Y , soon approaches a limiting curve (whose position depends slightly on $|\Delta E_g|$). For $|\Delta E_g| \approx 400 \text{ cm}^{-1}$, this curve has the form $Y \approx [1 + (E/32 \text{ cm}^{-1})^{2.5}]^{-1}$. Thermally averaging this with $W_a \propto E^{2.5}$ [see Eq. (16)] leads to a contribution $[1 + (T/28 \text{ K})^{2.5}]^{-1}$ to k/k_L . The summation over J , because of the cut-off at $J = J_1$, adds another factor which is given by the first term of Eq. (14). The two contributions together give a slightly stronger decrease of k/k_L with increasing temperature than is observed experimentally. Adjusting $|\Delta E_g|$ to $500 \pm 100 \text{ cm}^{-1}$ fits the data. Adding vibrational contributions would reproduce the turn-up of k at $T > 300$ K, such as demonstrated in ref. 7.

CONCLUSIONS

The present work has illustrated rotational effects in complex-forming bimolecular reactions which are characterized by a loose activated complex at the entrance and by a rigid activated complex at the exit. There can be "rotational channel switching" because of different rotational dependences of the entrance and exit threshold energies. Numbers of open channels governing the forward/backward dissociation ratios of the complex also depend on rotations in a marked way. Finally, for low temperatures, nuclear spin statistical weights of individual rotational levels should be taken into account.

These rotational effects have been demonstrated for the reaction $\text{CH}_4 + \text{O}_2^+$. The experimentally observed rate coefficient and its negative temperature coefficient can be reproduced in terms of rotational effects, if the lowering $|\Delta E_g|$ of the exit barrier compared with the entrance energy is treated as an adjustable parameter. Rotational channel switching here plays

an important role. $|\Delta E_g|$ must be larger than 300 cm^{-1} , otherwise lower rate coefficients would result than are observed experimentally. Because of the uncertainty in the detailed properties of the rigid activated complex (symmetry, rotational constants, vibrational frequencies), no unique prediction of the experiments would have been possible. However, the results can be rationalized satisfactorily. In summary, the present work emphasizes the crucial importance of rotational effects in the type of reactions considered, which has to be taken into account in addition to the generally considered vibrational effects.

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